Mai Nguyen

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Education

Colby College — Waterville, Maine

Major: Computer Science, Mathematics ***GPA: 4.03*** — *Expected Graduation: 2028*

Honors & Awards: Pulver Science Scholar, Second Prize in Vietnamese Mathematical Olympiad (VMO) 2024, Third Prize in Vietnamese Mathematical Olympiad (VMO) 2023

Leadership Experience: Founder of Colby Baking Club, Colby Outing Club Trip Leader, Colby Outdoor Orientation Trip Leader

Technical Skills

Languages: Python, JavaScript, C++, Java, C, OCaml
Frameworks & Libraries: React, Node.js, Express.js, Flask, scikit-learn, NumPy, pandas, JWT, Atomic Agents
Database & Cloud: MongoDB (NoSQL), Firebase, AWS S3
Tools & Technologies: Git/GitHub, Postman, Render, CORS, Rate Limiting, CI/CD, OpenAI GPT-4 API, Google Gemini API, ElevenLabs API

Experience

Undergraduate Research Assistant — Jun. 2026 – Present

McKelvey School of Engineering, Washington University in St. Louis — *St. Louis, MO*

- Developing a **polynomial-time** algorithm for guiding agent exploration in benign planning environments to efficiently learn an accurate environment model from trial-and-error
- Analyzed version space collapse using **combinatorial counting**, **pigeonhole arguments**, and **majority vote** elimination
- Evaluated **MCMC** and **DNF** approximate counting with provable error bounds

Machine Learning Research Assistant — Feb. 2025 – May. 2026

Department of Computer Science, Colby College — *Waterville, ME*

- Developed algorithms to incorporate personalized knowledge from user at test time to improve model accuracy on **5 mini dataset** to classify cooking recipe origin
- Evaluated predictive models (scikit-learn, NumPy, pandas) on survey data to simulate human-input scenarios
- Achieved near-ceiling accuracy **using only 50%** of available human-annotated features, slashing run time by 40%
- Presented findings at a school-wide research symposium to **200+ professors** and students [**Abstract**]

Research Assistant — Jun. – Aug. 2023

Project in Mathematics & Applications — *Vietnam*

- Built Statistics model for **Bulk RNA-Seq Quantification** using 3 ML models: Bayesian networks, Expectation-Maximization algorithm and Lagrange multiplier
- Processed data from single-end and paired-end RNA-Seq database using Python libraries
- Achieved **81% accuracy rate** in mRNA proportion prediction compared to industry-standard model after 30 trials

Projects

Can an RL Agent Beat Black-Scholes? – Options Hedging — Jun. 2026 – Present

- Developing a **PPO**-based RL agent to outperform **Black-Scholes** delta hedging under realistic transaction costs, framed as a **CVaR** minimisation problem over a custom **gymnasium** environment
- Designed a hedging **MDP** with continuous action space over GBM-simulated price paths, implementing full **P&L accounting** with transaction cost penalties and terminal option payoff
- Stress-testing agent robustness under **Heston stochastic volatility** and **Merton jump-diffusion** dynamics to evaluate out-of-distribution generalisation

Postcard I Never Send – Travel Storybook 🔗 — Feb. 2026 – Mar. 2026

- Built a full-stack web app with React and Node.js/Express that converts travel photos and trip details into narrative storybooks using **Google Gemini 2.5 Flash** with Google Search grounding
- Engineered multimodal AI pipeline to analyze uploaded photos against real-world place data, generating location-specific prose and captions
- Integrated **ElevenLabs TTS** narration with a custom audio player built from scratch using the Web Audio API
- Applied prompt engineering with strict output rules and style constraints to produce specific, human-sounding narratives

gastronome - not your grandma's recipe book 🔗 — Jul. 2025 – Present

- Built a cooking recipe ecosystem full-stack app with Node.js/Express, React, and Flask, deployed on Render with Firebase Authentication and AWS S3 integration
- Developed **20+ RESTful API** on recipes, users, and collections, with role-based access control
- Engineered AI pipeline using GPT-4 and Atomic Agents to extract and parse recipes from online sources
- Designed MongoDB schema with compound indices for optimal filtering across **1K+ recipe entries**
- Implemented a ML recommendation system that analyzes users' collections to produce personalized suggestions
- Deployed multi-service architecture on Render with rate limiting, CORS configuration, and image processing pipeline (AWS S3)